

MOAYEDPOUR Quentin

ENSAE 2nd year
Application Internship
2023-2024

Summary Report

ENSTA, Laboratoire U2IS

Palaiseau, France

Internship Supervisor : Gianni
FRANCHI

June 2024 - September 2024



ENSAE Paris

TSA 26644

Service des relations entreprises et des stages

5, avenue Henry Le Chatelier - 91764 PALAISEAU CEDEX - FRANCE

Tél : +33 (0)1 70 26 67 39 - Courriel : stage@ensae.fr

www.ensae.fr

Introduction

Anomaly detection in time series data is a crucial technique used to identify unusual or unexpected behaviors in data collected over time. This method is essential in various fields, such as finance, healthcare, and industrial monitoring, as it enables the rapid detection of potential issues, fraud, or equipment failures. By analyzing variations and patterns in time-series data, anomaly detection algorithms can alert users to abnormal events, aiding in decision-making. However, training a model involves numerous challenges, such as computational capabilities, the availability of extensive data, and data annotation, which motivates the development of transfer learning methods. The goal is to leverage a model that has been pre-trained on auxiliary data and adapt it to a specific task.

Deep Learning for Anomaly Detection

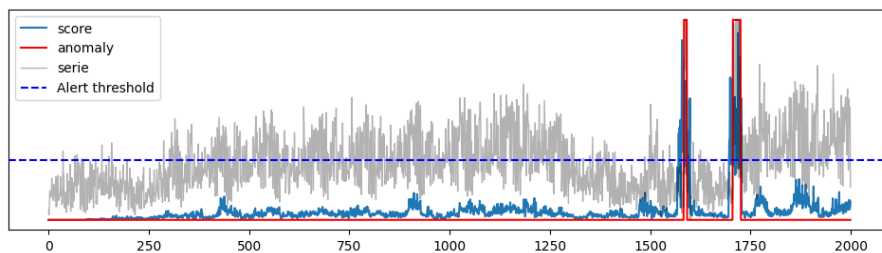
The development of deep learning methods has led to numerous models for anomaly detection. In the field of time series, three main approaches can be identified: anomaly detection through prediction error, reconstruction error, and representation. We focus particularly on **representation models** for anomaly detection. Given a time series in which potential anomalies are to be detected, the objective is to train and apply a model to extract relevant information associated with the series, facilitating anomaly detection. Since anomalies are generally rare, prior knowledge of their form is often unavailable. Neural networks, with their capacity to model complex relationships and learn relevant features, offer new perspectives for anomaly detection, contrasting with traditional statistical methods, which usually require strong assumptions.

Transfer Learning Algorithms

The goal of this project was to study various methods for detecting anomalies using **transfer learning** techniques. These methods are very popular for processing textual data and images but are still underutilized in time-series analysis.

An initial method explored was to use pre-trained models for prediction and derive an anomaly detector from the model's prediction error. The idea is that anomalies exhibit abnormal behavior in the series and are therefore more difficult for a prediction model to forecast. This method showed good results but did not outperform a statistical model (ARMA). We then sought to exploit the **latent representation** of a pre-trained model. A neural network

can be described as a sequential chain of numerous relatively "simple" operations, effectively solving a complex task through training on a large dataset. Data undergoes multiple transformations within a neural network, and a latent representation is an intermediate transformation within the model. Our proposed method uses a pre-trained model to represent time series data effectively in its latent space, applying it to a time series and learning an **"average representation"** of the serie in latent space. The idea is that a neural network learn how to extract meaning full information from the data, so we use a pre-train one that works as a feature extractor. We learn an average representation because if a segment of the series shows anomalies, its representation in latent space will likely differ from usual data. The algorithm select the meaning-full informations from the latent space and adjust some parameters in order to compute for each point of the series its distance to the average representation. We highly rely on the fact that anomalies are rare, so they'll have a limited impact on the learned parameters. The figure below shows an example of the model applied to a series. The model assigns an anomaly score, and a threshold is defined for alerts: if the score exceeds this threshold, the point is likely an anomaly.



Results and Conclusion

Our method, called "Moving Mahalanobis," demonstrated the best average results (compared to other tested methods) across a large number of datasets. The method also requires minimal computational resources and time to be applied. Since it uses a pre-trained model, it is easily applicable even for very long time series. The next step is to test the model with more different pre-trained models and to study hyperparameters to refine the model. Furthermore, we could think of adapting the algorithm for multivariate series: when for each time stamp, we have multiple points (For instance, financial markets are characterized by the prices of multiple assets).